



EUNOIA JUNIOR COLLEGE  
JC2 Preliminary Examination 2018  
General Certificate of Education Advanced Level  
Higher 1

CANDIDATE  
NAME

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CIVICS  
GROUP

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|---|---|---|--|--|
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INDEX  
NUMBER

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## CHEMISTRY

**8873/02**

Paper 2 Structured Questions

**13 September 2018**

**2 hours**

Candidates answer **Section A** and **Section B** on the Question Paper

Additional Materials: Data Booklet

### READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use an HB pencil for any diagrams or graphs.

Do not use paper clips, highlighters, glue or correction fluid.

#### Section A

Answer **all** questions in the spaces provided on the Question paper.

#### Section B

Answer **one** question in the spaces provided on the Question paper.

The use of an approved scientific calculator is expected, where appropriate.  
A Data Booklet is provided.

At the end of the examination, fasten all your work securely together.  
The number of marks is given in brackets [ ] at the end of each question or part question..

| For Examiner's Use   |  |
|----------------------|--|
| Section A (60 marks) |  |
| 1                    |  |
| 2                    |  |
| 3                    |  |
| 4                    |  |
| 5                    |  |
| Section B (20 marks) |  |
|                      |  |
| Total                |  |

This document consists of 22 printed pages.

## Section A

Answer **all** the questions in this section.

For  
Examiner's  
Use

- 1 Sulfur, atomic number 16, is found within the Earth's crust. Sulfur is released into the atmosphere at times of volcanic activity.

A sample of sulfur from a volcano was analysed to give the following composition of isotopes.

| isotope         | abundance (%) |
|-----------------|---------------|
| $^{32}\text{S}$ | 95.02         |
| $^{33}\text{S}$ | 0.76          |
| $^{34}\text{S}$ | 4.22          |

- (a) Define the term relative atomic mass.

Average / weighted / mean mass of the an atom of an element compared to

1/12th the mass of a  $^{12}\text{C}$  [1m, for both] (atom) or

The mass of 1 mole of an element compared to 1/12<sup>th</sup> the mass of 1 mole of  $^{12}\text{C}$

[1]

- (b) Calculate the relative atomic mass of the sample of sulfur.

$$\begin{aligned} \text{relative atomic mass of sulfur} &= \frac{(95.02 \times 32) + (0.76 \times 33) + (4.22 \times 34)}{100} \\ &= 32.092 \\ &\approx \underline{32.1} \text{ (to 1 d.p.) [1m]} \end{aligned}$$

[1]

- (c) John Dalton, an early 19th century scientist, believed that elements were made up of tiny particles called atoms which could not be divided. Nowadays, chemists know of the existence of sub-atomic particles in atoms and in ions.

Complete the table to show the number of sub-atomic particles in and the electronic configuration of the  $^{33}\text{S}$  atom and  $^{34}\text{S}^{2-}$  ion.

| isotopic species     | number of protons | number of neutrons | number of electrons | electronic configuration              |
|----------------------|-------------------|--------------------|---------------------|---------------------------------------|
| $^{33}\text{S}$      | 16                | 17                 | 16                  | $1s^2 \dots 2s^2 2p^6 3s^2 3p^4$ [1m] |
| $^{34}\text{S}^{2-}$ | 16                | 18                 | 18                  | $1s^2 \dots 2s^2 2p^6 3s^2 3p^6$ [1m] |

[4]

- (d) Solid sulfur exists as a lattice of  $S_8$  molecules. Each  $S_8$  molecule is a ring of eight atoms.

How many atoms of sulfur are there in 0.0120 mol of  $S_8$  molecules?

$$\begin{aligned}\text{number of sulfur atoms} &= n_{S_8} \times 8 \times N_A \\ &= 0.0120 \times 8 \times (6.02 \times 10^{23}) \\ &= 5.7792 \times 10^{22} \approx \underline{5.78 \times 10^{22}} \quad [1\text{m}]\end{aligned}$$

[1]

- (e) The only intermolecular forces in solid sulfur are instantaneous dipole-induced dipole.

- (i) Describe how instantaneous dipole-induced dipole arise.

Due to uneven movement of electrons or distribution of electrons [1m] in a molecule, which results in an instantaneous dipole / temporary dipole (in the molecule), which induces dipoles in neighbouring molecules. [1m, for both]

[2]

- (ii) Suggest why there are no other intermolecular forces in solid sulfur.

Only one type of atom or no (permanent) dipoles or non-polar / no polar bonds. [1m]

[1]

- (f) Sulfur hexafluoride,  $SF_6$ , exists as non-polar covalent molecules with an octahedral shape.

Explain why a molecule of  $SF_6$  has an octahedral shape.

The central sulfur has six  $\sigma$  bond pairs and no lone pairs [1m, for both] the bond pairs repel one another equally to minimise repulsion, resulting in an octahedral shape [1m]

[2]

[Total: 12]

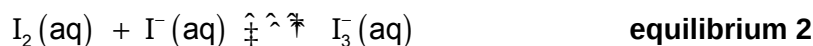
2 Iodine,  $I_2$ , is a grey-black solid that is not very soluble in water.

**Equilibrium 1** is set up with the equilibrium position well to the left.



Solid iodine is much more soluble in an aqueous solution of potassium iodide,  $KI(aq)$ , than in water.

**Equilibrium 2** is set up.



(a) Suggest why  $I_2$  is **not** very soluble in water.

Iodine is non-polar or Iodine is not able to form H bonds with water. [1m]  
.....[1]

(b) (i) Write an expression for the equilibrium constant for this reaction,  $K_c$ , stating its units.

$$K_c = \frac{[I_3^-]}{[I_2][I^-]} \quad [1m] ; \text{units: } \underline{\text{mol}^{-1} \text{ dm}^3} \quad [1m] \quad [2]$$

When 2.54 g of solid iodine is dissolved in 100  $\text{cm}^3$  of 1.00  $\text{mol dm}^{-3}$   $KI$ , and allowed to reach equilibrium, the resulting  $[I_3^-(aq)] = 9.98 \times 10^{-2} \text{ mol dm}^{-3}$ .

(ii) Calculate the equilibrium values of  $[I_2(aq)]$  and  $[I^-(aq)]$  in this solution and hence calculate a value for  $K_c$ .

|                                            | $I_2(aq)$                                               | + | $I^-(aq)$              | $I_3^-(aq)$            |
|--------------------------------------------|---------------------------------------------------------|---|------------------------|------------------------|
| initial conc.<br>/mol $\text{dm}^{-3}$     | $\frac{2.54}{253.8} \times \frac{100}{1000}$<br>= 0.100 |   | 1.00                   | 0                      |
| change in conc.<br>/mol $\text{dm}^{-3}$   | $-9.98 \times 10^{-2}$                                  |   | $-9.98 \times 10^{-2}$ | $+9.98 \times 10^{-2}$ |
| equilibrium conc.<br>/mol $\text{dm}^{-3}$ | $2.00 \times 10^{-4}$                                   |   | 0.9002                 | $9.98 \times 10^{-2}$  |

$$K_c = \frac{[I_3^-]}{[I_2][I^-]} = \frac{9.98 \times 10^{-2}}{2.00 \times 10^{-4} \times 0.9002} = 554.321 \approx \underline{554} \text{ mol}^{-1} \text{ dm}^3$$

$$[I_2(aq)] = \underline{2.00 \times 10^{-4} \text{ mol dm}^{-3}} \quad [1m]$$

$$[I^-(aq)] = \underline{0.900 \text{ mol dm}^{-3}} \quad [1m]$$

$$K_c = \underline{554 \text{ mol}^{-1} \text{ dm}^3} \quad [1m] \quad [3]$$

- (c) The student adds an excess of aqueous silver nitrate,  $\text{AgNO}_3(\text{aq})$ , to the equilibrium mixture forming a yellow  $\text{AgI}$  precipitate.

State what other observation will be seen and explain the observations in terms of both **equilibrium 1** and **equilibrium 2** and any species formed.

Precipitation of  $\text{AgI}$  causes  $[\text{I}^-]$  to **decrease**. This causes **equilibrium 2** to shift to the **left**, which in turn causes  $[\text{I}_2(\text{aq})]$  to **increase**. Increase in  $[\text{I}_2(\text{aq})]$  causes **equilibrium 1** to shift to **left**. **[1m] for all**

Hence  $\text{I}_2(\text{s})$  comes out of solution as a **black** or grey **precipitate**. **[1m]**

.....[2]

- (d) Two redox reactions of iodine are described below.

**Reaction 1:** Iodine is reacted with oxygen to form a compound with a molar mass of  $333.8 \text{ g mol}^{-1}$ .

**Reaction 2:** Under alkaline condition, iodine disproportionates to form iodide ions and iodate(V),  $\text{IO}_3^-$  ions.

Construct equations for these **two** reactions.

Reaction 1:

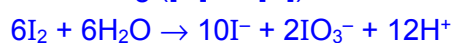


Reaction 2:

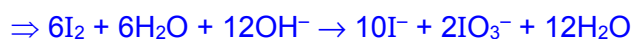


[Total: 10]

Combining ([O] + 5[R]):



Add  $12\text{OH}^-$  to both sides:



- 3 Table 3.1 gives some data on four fuel sources: methanol, ethanol, hydrogen and octane. Octane can serve as a rough approximation of petrol.

For  
Examiner's  
Use

Table 3.1

| compound        | formula                          | molar mass<br>/ g mol <sup>-1</sup> | density<br>/ g cm <sup>-3</sup> | $\Delta H_c^{\ddagger}$ (298 K)<br>/ kJ mol <sup>-1</sup> | $\Delta H_f^{\ddagger}$ (298 K)<br>/ kJ mol <sup>-1</sup> |
|-----------------|----------------------------------|-------------------------------------|---------------------------------|-----------------------------------------------------------|-----------------------------------------------------------|
| methanol        | CH <sub>3</sub> OH               | 32.0                                | 0.793 <sup>a</sup>              | -726.0                                                    | -239.1                                                    |
| ethanol         | C <sub>2</sub> H <sub>5</sub> OH | 46.0                                | 0.789 <sup>a</sup>              | -1367.3                                                   | -277.1                                                    |
| liquid hydrogen | H <sub>2</sub>                   | 2.0                                 | 0.0711 <sup>b</sup>             |                                                           |                                                           |
| octane          | C <sub>8</sub> H <sub>18</sub>   | 114.0                               | 0.703 <sup>a</sup>              |                                                           | -250.0                                                    |

<sup>a</sup> At 298 K and 1 bar pressure

<sup>b</sup> At 20 K and 1 bar pressure

- (a) Insert the missing molar mass values in the Table 3.1. [1]

- (b) Calculate the density of **gaseous** hydrogen at 298 K and 1 bar pressure. Assume 1 mol of any gas occupies 24 dm<sup>3</sup> at 298 K and 1 bar pressure. Give your answer in g cm<sup>-3</sup>.

$$\text{density} = \frac{\text{molar mass}}{\text{molar volume}} = \frac{2 \text{ g mol}^{-1}}{24 \text{ dm}^3 \text{ mol}^{-1}} = \frac{2 \text{ g mol}^{-1}}{24000 \text{ cm}^3 \text{ mol}^{-1}} \\ = 8.33 \times 10^{-5} \text{ g cm}^{-3}$$

density =  $8.33 \times 10^{-5}$  [1m] g cm<sup>-3</sup> [1]

- (c) What is the value of the standard enthalpy of formation of hydrogen **gas**, H<sub>2</sub>?

0 kJ mol<sup>-1</sup> [1m] [1]

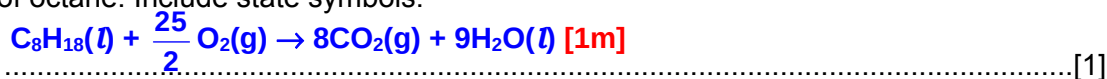
- (d) Use the information in Table 3.2 to give the value of the standard enthalpy of combustion of hydrogen at 298 K.

Table 3.2

| compound       | $\Delta H_f^{\ddagger}$ (298 K) / kJ mol <sup>-1</sup> |
|----------------|--------------------------------------------------------|
| water          | -285.8                                                 |
| carbon dioxide | -393.5                                                 |

$\Delta H_c^{\ddagger}$  of hydrogen at 298 K =  $-285.8$  [1m] kJ mol<sup>-1</sup> [1]

- (e) Write down the chemical equation that represents the *standard enthalpy of combustion* of octane. Include state symbols.



- (f) Use the enthalpy of formation data in Table 3.1 and Table 3.2 and the equation in (e) to calculate the standard enthalpy of combustion,  $\Delta H_c^\ominus$ , of octane.

$$\begin{aligned} \Delta H_c^\ominus (\text{octane}) &= \sum \Delta H_f^\ominus (\text{products}) - \sum \Delta H_f^\ominus (\text{reactants}) \\ &= 9\Delta H_f^\ominus (\text{H}_2\text{O}) + 8\Delta H_f^\ominus (\text{CO}_2) - \Delta H_f^\ominus (\text{C}_8\text{H}_{18}) - \frac{25}{2} \Delta H_f^\ominus (\text{O}_2) \\ &= 9(-285.8) + 8(-393.5) - (-250.0) - 0 \\ &= -5470.2 \approx \underline{-5470 \text{ kJ mol}^{-1}} \quad [1\text{m}] \end{aligned}$$

[2]

- (g) An important property of a fuel, especially when the fuel has to be lifted (such as in aviation), is the energy released on combustion *per gram* of fuel.

Calculate the enthalpy change of combustion per gram of fuel at 1 bar pressure and 298 K for methanol and hydrogen gas.

- (i) methanol

$$\Delta H_c^\ominus (\text{kJ g}^{-1}) = \frac{\Delta H_c^\ominus (\text{kJ mol}^{-1})}{\text{molar mass}} = \frac{-726 \text{ kJ mol}^{-1}}{32.0 \text{ g mol}^{-1}} = -22.6875 \approx \underline{-22.7 \text{ kJ g}^{-1}}$$

$$\Delta H_c^\ominus = \underline{-22.7 [1\text{m}]} \text{ kJ g}^{-1} [1]$$

- (ii) hydrogen gas

$$\Delta H_c^\ominus (\text{kJ g}^{-1}) = \frac{\Delta H_c^\ominus (\text{kJ mol}^{-1})}{\text{molar mass}} = \frac{-285.8 \text{ kJ mol}^{-1}}{2.0 \text{ g mol}^{-1}} = -142.9 \approx \underline{-143 \text{ kJ g}^{-1}}$$

$$\Delta H_c^\ominus = \underline{-143 [1\text{m}]} \text{ kJ g}^{-1} [1]$$

- (h) Another important characteristic of a fuel, especially when there is a fuel tank of limited size, is the energy released on combustion *per cm<sup>3</sup>* of fuel.

Calculate the enthalpy change of combustion per cm<sup>3</sup> of fuel for ethanol and octane.

- (i) ethanol

$$\Delta H_c^\ominus (\text{kJ cm}^{-3}) = \frac{\Delta H_c^\ominus (\text{kJ mol}^{-1})}{\text{molar mass}} \times \text{density} = \frac{-1367.3 \text{ kJ mol}^{-1}}{46.0 \text{ g mol}^{-1}} \times 0.789 \text{ g cm}^{-3}$$

$$= -23.452 \approx \mathbf{-23.5 \text{ kJ cm}^{-3}}$$

$$\Delta H_c^\ominus = \dots\dots\dots \mathbf{-23.5 [1m]} \dots\dots\dots \text{kJ cm}^{-3} [1]$$

- (ii) octane

$$\Delta H_c^\ominus (\text{kJ cm}^{-3}) = \frac{\Delta H_c^\ominus (\text{kJ mol}^{-1})}{\text{molar mass}} \times \text{density} = \frac{-5470.2 \text{ kJ mol}^{-1}}{114.0 \text{ g mol}^{-1}} \times 0.703 \text{ g cm}^{-3}$$

$$= -33.7329 \approx \mathbf{-33.7 \text{ kJ cm}^{-3}}$$

$$\Delta H_c^\ominus = \dots\dots\dots \mathbf{-33.7 [1m]} \dots\dots\dots \text{kJ cm}^{-3} [1]$$

- (i) Explain why, given the data in the question, it is not strictly possible to make a fair comparison of the energy released per cm<sup>3</sup> of liquid hydrogen with the other fuels.

The enthalpy change of combustion value for hydrogen is for 298 K, and so relates to gaseous hydrogen, not to liquid hydrogen / no account taken of different temperatures / latent heat of vaporisation of hydrogen [1m].....[1]

[Total: 12]



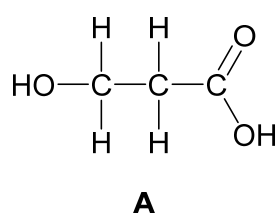
- 4 (a) 3-Hydroxypropanoic acid,  $\text{HOCH}_2\text{CH}_2\text{COOH}$ , can be produced microbiologically from sugars in corn.  $\text{HOCH}_2\text{CH}_2\text{COOH}$  can be used as a 'green' starting material for the synthesis of many organic compounds including some important polymers.

Three synthetic routes are shown below for converting  $\text{HOCH}_2\text{CH}_2\text{COOH}$  (**A**) into different polymers.

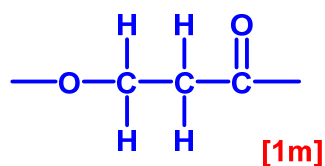
The names of the processes for each synthetic step are given.

- (i) In the boxes below, give the structures of the organic compounds formed.

**Synthesis 1**

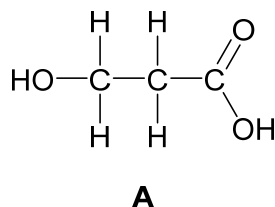


polymerisation

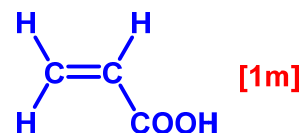


one repeat unit

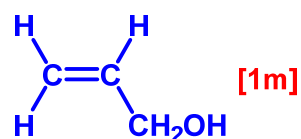
**Synthesis 2**



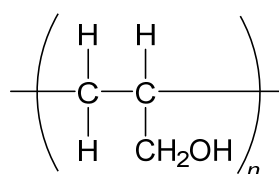
elimination



reduction



polymerisation



PVA, used in adhesives

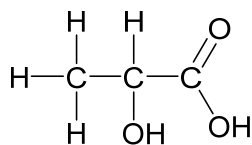
### Synthesis 3



- [1]

- $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$ ,  $\text{H}_2\text{SO}_4(\text{aq})$ , heat with reflux [1m]** [1]

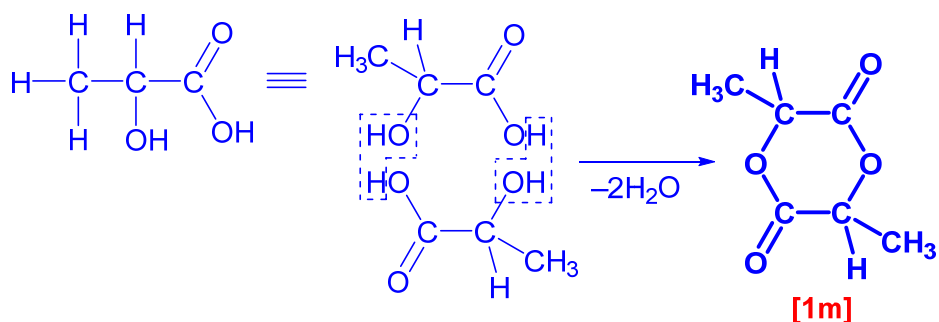
(b) 2-Hydroxypropanoic acid, also known as lactic acid, has structure shown below.



(i) When heated strongly, lactic acid forms a cyclic 'diester'.

The diester has the molecular formula,  $\text{C}_6\text{H}_8\text{O}_4$ .

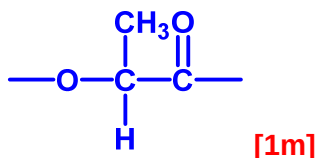
Draw the structure of the cyclic diester.



[1]

(ii) Poly(lactic acid), PLA, is used to make 'dissolvable' stitches (for holding wounds together). PLA breaks down into smaller molecules after one or two weeks.

Draw the structure of **one** repeat unit in PLA.



[1m]

[1]

(iii) Explain how PLA breaks down and why the stitches 'dissolve'.

Ester linkages in PLA are hydrolysed to regenerate lactic acid [1m]

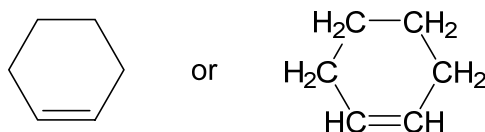
The lactic acid monomer is soluble in water because it forms hydrogen

bonds to water [1m]

[2]

[Total: 13]

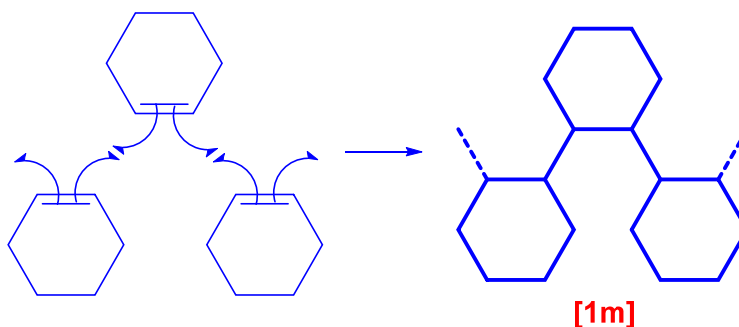
5 Cyclohexene behaves as a typical alkene.



(a) (i) Give the name of the type of polymerisation that cyclohexene undergoes.

Addition polymerisation [1m].....[1]

(ii) Draw a section of the polymer consisting of **three** repeat units.



Three monomer units lacking a double bond joined together correctly;  
dotted bonds at the ends [1]

(iii) A sample of cyclohexene is polymerised to a relative molecular mass of 2500, on average.

Calculate the number of complete cyclohexene units that polymerise in each polymer molecule on average. Show your working.

$$\text{number of complete cyclohexene units} = \frac{2500}{M_r(\text{C}_6\text{H}_{10})} = \frac{2500}{82} = 30.4878$$

≈ 30

number of cyclohexene units = ..... 30 [1m]..... [1]

(b) Cyclohexene will react with hydrogen.

(i) Write an equation for the reaction.

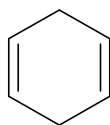


[1]

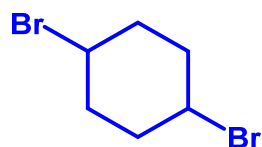
(ii) State the conditions required.

nickel catalyst at 10 atm (or high pressure) or platinum catalyst, r.t.p. [1m]...[1]

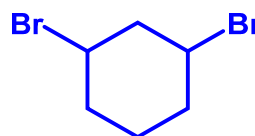
- (c) Cyclohexa-1,4-diene also displays reactivity typical of alkenes. Its structure is shown.



Draw the structures of all possible products of the reaction when one molecule of cyclohexa-1,4-diene completely reacts with two molecules of **hydrogen bromide**.



[1m] each



[2]

- (d) A graphene sheet is a layer of graphite.

A recent development has been the synthesis of graphene ribbons (reported in Nature, 2010). A reaction scheme is shown Fig. 5.1.

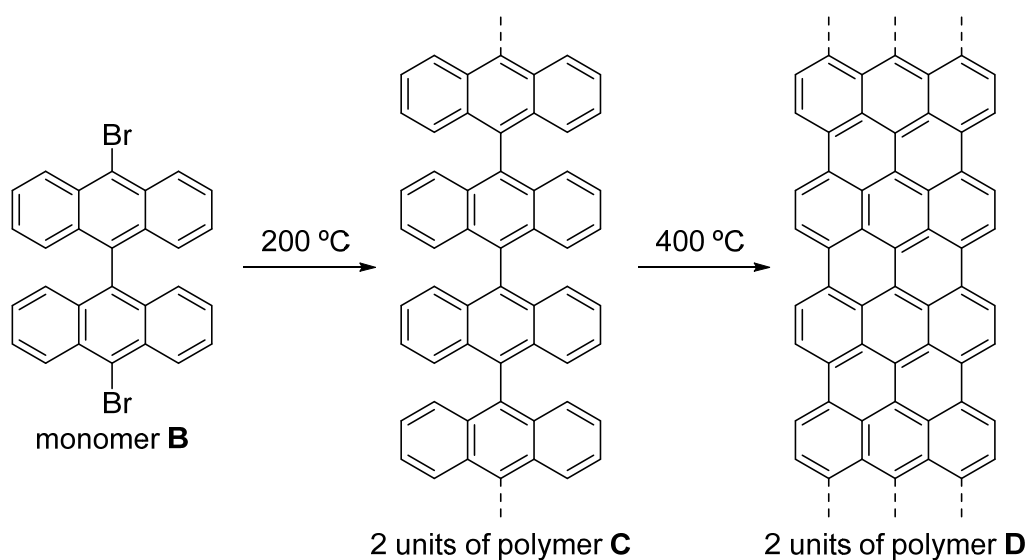


Fig. 5.1

- (i) When monomer **B** is polymerised to make **C** there is also another product, **X**.

Give the molecular formula of **X**.

**X** is .....  $\text{Br}_2$  [1m] .....

[1]

- (ii) In the transformation of polymer **C** into polymer **D**, another product, **Y**, is produced.

Give the molecular formula of **Y**.

**Y** is .....  $\text{H}_2$  [1m] ..... [1]

- (iii) Deduce the number of moles of **X** and **Y** produced per mole of monomer **B**.

number of moles of **X** = ..... 1 [1m] .....

number of moles of **Y** = ..... 4 [1m] ..... [2]

- (iv) Boron nitride, BN, forms sheets similar to graphene except they contain dative covalent bonds as well as covalent bonds.

Add all the possible **dative** covalent bonds between the atoms shown in the structure Fig. 5.2.

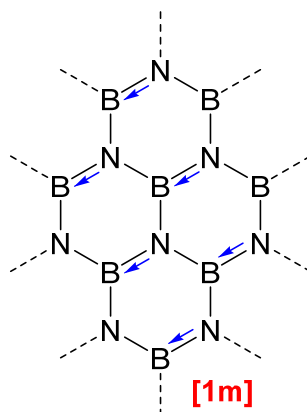


Fig. 5.2

[1]

- (v) Boron nitride can also form a giant covalent structure in which each atom has four single bonds.

Suggest the name of another substance which has this type of structure.

diamond or silicon or germanium [1m] ..... [1]

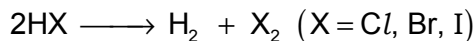
[Total: 13]

## Section B

Answer **one** question in this section, in the space provided.

For  
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- 6 (a) Under certain conditions the halogen halides decompose into their elements.



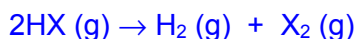
- (i) How can this reaction be carried out?

Insert a hot wire / glass rod into a test-tube of the gas. [1m].....[1]

- (ii) Describe what will be observed in the case of HI.

Purple/violet vapour of I<sub>2</sub> will be observed. [1m].....  
.....[1]

- (iii) Use bond energy values from the *Data Booklet* to calculate  $\Delta H$  for this reaction when X = Cl and when X = I.



$$\begin{aligned} \Delta H &= \sum \text{BE}(\text{bonds broken}) - \sum \text{BE}(\text{bonds formed}) \\ &= 2 \times \text{BE}(\text{H}-\text{X}) - \text{BE}(\text{H}-\text{H}) - \text{BE}(\text{X}-\text{X}) \end{aligned}$$

For X = Cl      $\Delta H = 2 \times 431 - 436 - 244 = \underline{+182 \text{ kJ mol}^{-1}} \text{ [1m]}$

For X = I:      $\Delta H = 2 \times 299 - 436 - 151 = \underline{+11 \text{ kJ mol}^{-1}} \text{ [1m]}$

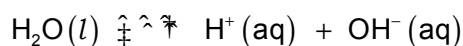
[2]

- (iv) Explain how the extent of decomposition of the hydrogen halides vary down Group 17 using the results from (a)(iii).

The results from (a)(iii) shows that down the group, the reaction becomes less  
endothermic (or less energy is required to carry out the reaction) [1m].  
therefore the extent of decomposition of the hydrogen halides increases down  
the group. [1m]

[2]

- (b) The dissociation of water is a reversible reaction.



The ionic product of water,  $K_w$ , measures the extent of dissociation of water.

$K_w$  varies with temperature. Therefore, it is always important to quote the temperature at which measurements are being taken.

Fig. 6.1 shows the variation of  $K_w$  between 0 °C and 60 °C.

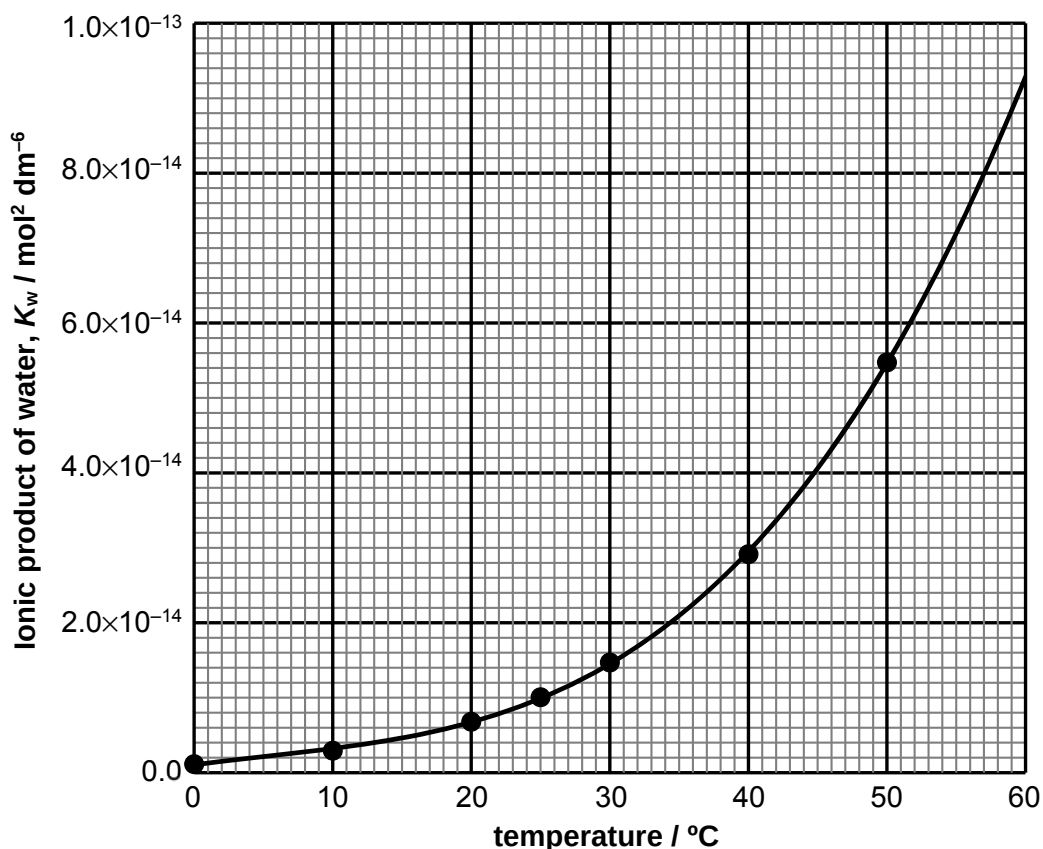


Fig. 6.1

- (i) Write the expression for  $K_w$ .

$$K_w = [\text{H}^+(\text{aq})][\text{OH}^-(\text{aq})] \quad [1\text{m}] \dots\dots\dots [1]$$

- (ii) Calculate the  $\text{OH}^-(\text{aq})$  concentration in an aqueous solution of hydrochloric acid with a pH of 4.37 at 25 °C.

$$[\text{H}^+] = 10^{-\text{pH}} = 10^{-4.37} = 4.266 \times 10^{-5} \text{ mol dm}^{-3} \quad [1\text{m}]$$

$$[\text{OH}^-] = \frac{K_w}{[\text{H}^+]} = \frac{1.0 \times 10^{-14}}{4.266 \times 10^{-5}} = 2.344 \times 10^{-10} \approx \underline{2.34 \times 10^{-10} \text{ mol dm}^{-3}} \quad [1\text{m}]$$

[2]



- (iii) Using Fig. 6.1, explain whether the dissociation of water is an exothermic or endothermic process.

When temperature increases,  $K_w$  increases, i.e. equilibrium shift right to absorb the heat. Hence the dissociation of water is endothermic [1m]

.....[1]

- (iv) Determine the pH of pure water at body temperature, 37 °C.

From Fig. 6.1, at 37 °C,  $K_w = 2.4 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$  [1m]

$$K_w = [\text{H}^+][\text{OH}^-] = 2.4 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$$

$$\text{Since } [\text{H}^+] = [\text{OH}^-],$$

$$[\text{H}^+]^2 = 2.4 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$$

$$[\text{H}^+] = \sqrt{2.4 \times 10^{-14}} = 1.5492 \times 10^{-7} \text{ mol dm}^{-3} \text{ [1m]}$$

$$\text{pH} = -\lg[\text{H}^+] = -\lg(1.5492 \times 10^{-7}) = 6.8099 \approx 6.81 \text{ [1m]}$$

[3]

- (c) In humans it is important for blood to be maintained at a pH between 7.35 and 7.45. One of the way it does this is by using a buffer of  $\text{CO}_2(\text{aq})$  and  $\text{HCO}_3^-(\text{aq})$ .

During vigorous exercise the muscles produce lactic acid. Lactic acid is transported by the blood to the liver to be broken down.

- (i) State the effect that exercise will have on the pH of blood if there is no buffer present.

Since lactic acid is produced, the pH of blood will decrease [1m].....[1]

- (ii) Write equation(s) to explain how the  $\text{CO}_2(\text{aq}) / \text{HCO}_3^-(\text{aq})$  buffer system helps to maintain the pH of blood between 7.35 and 7.45 during exercise.

The conjugate base,  $\text{HCO}_3^-$ , will react with  $\text{H}^+$  from the lactic acid:



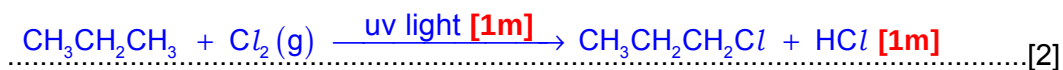
removing the lactic acid [1m] produced in the muscle, thus maintaining the pH of blood.

.....[2]

- (c) Chlorine reacts with both propane and propene, but under different conditions.

Write equations, giving the structural formulae of the organic products, and state the conditions for the reaction of chlorine.

- (i) with propane



- (ii) with propene

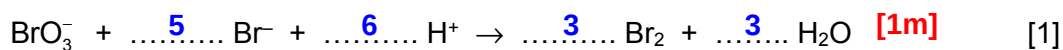


[Total: 20]

- 7 (a) In the presence of acid,  $\text{H}^+(\text{aq})$ , aqueous bromate(V) ions,  $\text{BrO}_3^-(\text{aq})$ , react with aqueous bromide ions,  $\text{Br}^-(\text{aq})$ , to produce bromine,  $\text{Br}_2(\text{aq})$ .

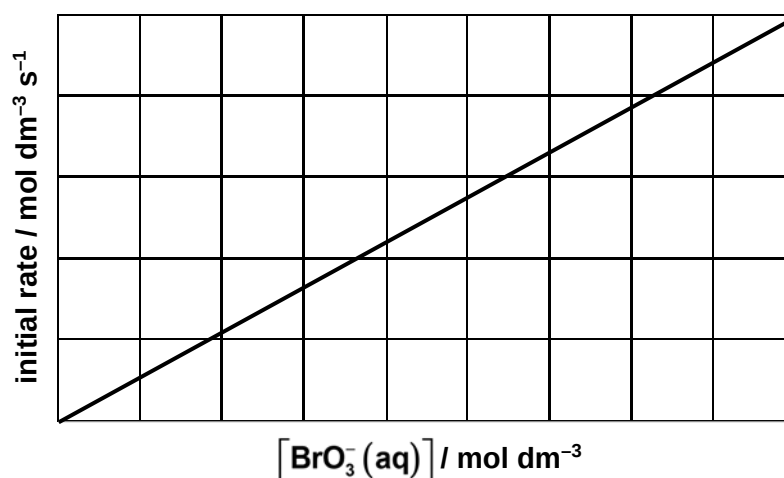
A student carried out an investigation into the kinetics of this reaction.

- (i) Balance the ionic equation for this reaction.



The student investigated how different concentrations of  $\text{BrO}_3^-(\text{aq})$  affect the initial rate of the reaction.

A graph of initial rate against  $[\text{BrO}_3^-(\text{aq})]$  is shown below.



The student then investigated how different concentrations of  $\text{Br}^-(\text{aq})$  and  $\text{H}^+(\text{aq})$  affect the initial rate of the reaction.

The results are shown below.

| $[\text{BrO}_3^-(\text{aq})]$<br>/ $\text{mol dm}^{-3}$ | $[\text{Br}^-(\text{aq})]$<br>/ $\text{mol dm}^{-3}$ | $[\text{H}^+(\text{aq})]$<br>/ $\text{mol dm}^{-3}$ | initial rate<br>/ $\text{mol dm}^{-3} \text{s}^{-1}$ |
|---------------------------------------------------------|------------------------------------------------------|-----------------------------------------------------|------------------------------------------------------|
| $5.0 \times 10^{-2}$                                    | $1.5 \times 10^{-1}$                                 | $3.1 \times 10^{-1}$                                | $1.19 \times 10^{-5}$                                |
| $5.0 \times 10^{-2}$                                    | $3.0 \times 10^{-1}$                                 | $3.1 \times 10^{-1}$                                | $2.38 \times 10^{-5}$                                |
| $5.0 \times 10^{-2}$                                    | $1.5 \times 10^{-1}$                                 | $6.2 \times 10^{-1}$                                | $4.76 \times 10^{-5}$                                |

- (ii) Using the results from the student's experiments, determine the order of reactions with respect to all the three reactants.

Graph of initial rate against  $[\text{BrO}_3^- (\text{aq})]$ :

Straight/diagonal line through origin thus is **1st order** with respect to  $\text{BrO}_3^-$ .

Using initial rates data:

When  $[\text{Br}^-]$  is doubled, rate  $\times 2$  thus **1st order** with respect to  $\text{Br}^-$

When  $[\text{H}^+] \times 2$ , rate  $\times 4$  ( $2^2$ ) thus **2nd order** with respect to  $\text{H}^+$

order w.r.t.  $\text{BrO}_3^-$  = ..... **1 [1m]**

order w.r.t.  $\text{Br}^-$  = ..... **1 [1m]**

order w.r.t.  $\text{H}^+$  = ..... **2 [1m]**.....[3]

- (iii) Write the rate equation and calculate the rate constant for this reaction, including the units.

Rate equation is rate =  $k[\text{BrO}_3^-][\text{Br}^-][\text{H}^+]^2$  **[1m]**

Using the first set of data,

$$k = \frac{\text{rate}}{[\text{BrO}_3^-][\text{Br}^-][\text{H}^+]^2} = \frac{1.19 \times 10^{-5}}{(5.0 \times 10^{-2})(1.5 \times 10^{-1})(3.1 \times 10^{-1})^2}$$

$$= 0.01651 \approx \underline{\underline{0.0165}} \text{ [1m] mol}^{-3} \text{ dm}^9 \text{ s}^{-1} \text{ [1m]}$$

[3]

- (b) Sodium, magnesium, aluminium, silicon, phosphorus, sulfur, chlorine and neon are elements in period 3 of the periodic table.

- (i) State and explain the differences between the ionic radii of  $\text{Na}^+$ ,  $\text{Si}^{4+}$ , and  $\text{P}^{3-}$  and  $\text{Cl}^-$ .

The anions have larger ionic radii than the cations as the anions have an extra quantum shell of electrons. **[1m]**

$\text{Si}^{4+}$  and  $\text{Na}^+$  are isoelectronic, so are  $\text{P}^{3-}$  and  $\text{Cl}^-$ . But the nuclear charge of Si is higher than that of Na, and that of Cl is higher than that of P. Hence the effective nuclear charge of  $\text{Si}^{4+}$  is higher than  $\text{Na}^+$ , and that of  $\text{Cl}^-$  is higher than  $\text{P}^{3-}$ . Attraction of the electrons by the nucleus (protons) is greater, thus resulting in a smaller ionic radius for  $\text{Si}^{4+}$  and  $\text{Cl}^-$ . **[1m]**

[2]

- (ii) Sodium, silicon, phosphorus and chlorine also differ greatly in their melting points. Describe how their melting points differ, and explain this variation.

The melting points of  $\text{Si} > \text{Na} > \text{P}_4 > \text{Cl}_2$ . [1m]

Si has a giant covalent structure with strong covalent bonds between the atoms. [1m] A lot of energy is required to break the strong covalent bonds, hence the very high melting point.

Na has giant metallic structure. Metallic bonds are stronger than the dispersion forces present in  $\text{P}_4$  and  $\text{Cl}_2$  and Na has higher melting point than the 2 non-metals. However, as Na has only 1 valence electron per atom, so the metallic bond is much weaker than the covalent bonding in silicon. [1m]

Dispersion forces strength is proportional to the number of electrons in the molecules.  $\text{P}_4$  has more polarisable electron cloud than  $\text{Cl}_2$ , hence P has higher melting point. [1m] [4]

- (iii) The acid-base behaviour of aluminium oxide,  $\text{Al}_2\text{O}_3$ , shows similarities to that of magnesium oxide,  $\text{MgO}$ , on the one hand, and phosphorus(V) oxide,  $\text{P}_4\text{O}_{10}$ , on the other.

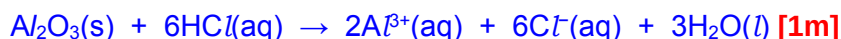
Describe what these similarities are, and explain why aluminium oxide occupies this in-between position.

Write equations for all the reactions you choose to illustrate your answer.

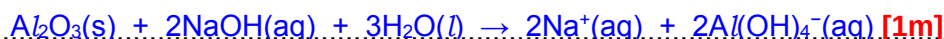
Aluminium oxide is both basic, like magnesium oxide, and acidic, like phosphorus(V) oxide. [1m] It is amphoteric ('in-between position'). It is an ionic oxide with some covalent character.

$\text{Al}^{3+}$  is a small and highly charged ion. Its high charge density gives it great polarising power. [1m] This results in the polarisation of the  $\text{O}^{2-}$  ion, giving the ionic bond between  $\text{Al}^{3+}$  and  $\text{O}^{2-}$  ions some covalent character. [1m]

$\text{Al}_2\text{O}_3$  shows its basic property when it reacts with acids, e.g.  $\text{HCl}$



$\text{Al}_2\text{O}_3$  shows its acidic property when it reacts with alkalis, e.g.  $\text{NaOH}$



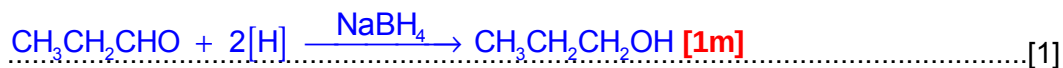
[5]

- (c) Sodium borohydride,  $\text{NaBH}_4$ , is a useful reducing agent in organic chemistry. It reduces both propanal and propanone.

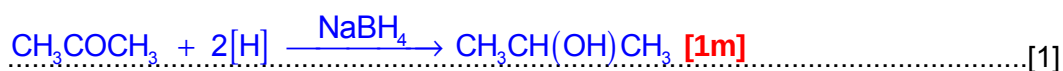
For  
Examiner's  
Use

Write balanced equations, giving the structural formulae of the organic products, to illustrate its reducing property.

- (i) with propanal



- (ii) with propanone



[Total: 20]